KEY FINANCIAL FORMULAS

Monte Carlo Portfolio Risk Simulator

Equations for portfolio simulation, outcome analysis and tail-risk measurement

1. Returns and Portfolio Construction

1. Daily Asset Return

P(i,t): asset price today; P(i,t−1): previous price; R(i,t): daily return.

$$R_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1$$

2. Portfolio Daily Return

R(p,t): portfolio return; w(i): portfolio weight; N: number of assets.

$$R_{p,t} = \sum_{i=1}^{N} w_i R_{i,t}$$

3. Equal-Weight Portfolio

Each asset receives the same allocation; all weights sum to one.

$$w_i = \frac{1}{N}, \qquad \sum_{i=1}^{N} w_i = 1$$

4. Covariance Matrix

Sigma represents the variance and covariance structure of asset returns.

$$\Sigma = \mathrm{Cov}(\mathbf{R})$$

2. Monte Carlo Simulation

5. Correlated Random Shocks

L: Cholesky factor; z: independent random vector; Sigma: covariance matrix.

$$\boldsymbol{\varepsilon}_t = L\mathbf{z}_t, \qquad LL^\top = \Sigma$$

6. Standard Normal Shocks

z is a vector of independent standard-normal random variables.

$$\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, I)$$

7. Terminal Portfolio Value

V(0): initial value; V(T): terminal value; T: simulated periods.

$$V_T = V_0 \prod_{t=1}^{T} (1 + R_{p,t})$$

8. Terminal Return

Percentage gain or loss relative to the initial portfolio value.

$$R_T = \frac{V_T}{V_0} - 1$$

3. Risk Metrics and Simulation Outcomes

9. Portfolio Loss

Positive L denotes a loss; negative L denotes a gain under this convention.

$$L = V_0 - V_T$$

10. Probability of Loss

M: number of simulations; indicator equals 1 when the condition is true.

$$\hat{P}(V_T < V_0) = \frac{1}{M} \sum_{j=1}^{M} \mathbf{1}(V_T^{(j)} < V_0)$$

11. Value at Risk

The alpha-quantile of the simulated loss distribution; verify the code's convention.

$$\mathrm{VaR}_\alpha = Q_\alpha(L)$$

12. Expected Shortfall

Average loss in the tail at or beyond the VaR threshold.

$$\mathrm{ES}_\alpha = \mathbb{E}[L \mid L \geq \mathrm{VaR}_\alpha]$$

13. Terminal Value Percentile

The alpha percentile is the terminal value at or below which that share of outcomes falls.

$$q_\alpha = Q_\alpha(V_T^{(1)}, \ldots, V_T^{(M)})$$

14. Mean Terminal Value

Arithmetic mean across all simulated terminal portfolio values.

$$\bar{V}_T = \frac{1}{M} \sum_{j=1}^{M} V_T^{(j)}$$

4. Historical Performance Reference

15. Drawdown

Decline from the running peak. Maximum drawdown is the most negative drawdown.

$$\mathrm{DD}_t = \frac{V_t}{\max_{s \le t} V_s} - 1$$

16. Compound Annual Growth Rate

Y: investment period in years; V(0): initial value; V(T): final value.

$$\mathrm{CAGR} = \left(\frac{V_T}{V_0}\right)^{1/Y} - 1$$

17. Annualized Volatility

Uses the conventional 252 trading-day annualization assumption.

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \sqrt{252}$$

18. Sharpe Ratio

Annualized excess return divided by annualized volatility.

$$S = \frac{R_{\text{annual}} - R_f}{\sigma_{\text{annual}}}$$

Definitions of Variables

Symbol	Meaning	Symbol	Meaning
R(i,t)	Return of asset i at time t	Σ	Covariance matrix
w(i)	Weight assigned to asset i	L	Loss; also used for Cholesky factor in matrix equation
V(0)	Initial portfolio value	V(T)	Terminal portfolio value
M	Number of simulation paths	α	Quantile/confidence level
VaR	Value at Risk	ES	Expected Shortfall
σ	Standard deviation / volatility	Y	Investment period in years

Methodology note: This is a formula reference for a correlated Monte Carlo portfolio-risk workflow. Verify the exact return-generation method, covariance treatment, VaR quantile convention, and tail selection against the project source code before describing them as implemented details. Simulated outcomes are not guarantees of future performance.

Educational project reference · Not investment advice